

Team Worksheet — Lab 8

Value Stream Mapping, Takt Time and the Eight Wastes — In Depth

Team members: _____

Date: _____

1. Our understanding of the problem

2. Our working

Use the data in `lab-08-value-stream-map.xlsx`.

3. What we produced

Deliverable: A current-state VSM with a value-added ratio, takt time and a quantified waste log.

4. What the evidence does NOT tell us

5. Our conclusion

6. Step notes

Step 1. Recap the VSM in full: what it shows that a process map does not — inventory, information flow, lead time and the push/pull signal.

Step 2. Recap the standard VSM icons: process box, data box, inventory triangle, push arrow, supermarket, truck, information flow and the timeline ladder.

Step 3. Map the current state of your capstone value stream from customer order to delivery. Include the information flow that triggers each step.

Step 4. Populate each data box: cycle time, changeover time, uptime, batch size, number of operators and first-pass yield.

Step 5. Draw the timeline ladder underneath. Put value-added time on the lower rung and waiting/queue time on the upper rung.

Step 6. Calculate total lead time, total value-added time, and the value-added ratio (VA time / lead time). Most processes land under 5 percent.

Step 7. Calculate takt time = available production time / customer demand for the period. Compare takt against each step's cycle time to find the bottleneck.

Step 8. Recap the eight wastes in full (DOWNTIME: Defects, Overproduction, Waiting, Non-utilised talent, Transport, Inventory, Motion, Extra processing).

Step 9. Walk your value stream and log every instance of each waste. Quantify each in time or money — an unquantified waste will not win funding.

Step 10. Mark the three points on the VSM where the largest waste concentrates. These become improvement candidates for Day 4.

7. Self-check

Your VSM shows both material and information flow, you have calculated the value-added ratio and takt time from real numbers, and every logged waste carries a time or dollar quantification.

Complete? ■ Yes ■ Not yet

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